

https://drive.google.com/file/d/1R6uIHjx683x15g9ifKENDlhmVrw_t1_c/view

★ AI:

- Study and design of intelligent agents, where an intelligent agent is a system that **perceives its environment** and **takes actions (thru actuators) that maximize chances of success**

★ applications of AI:

- **Smart search algos** : games, route finding, transportation
- **NLP applications**: search engines, OCR, speech recognition, spelling checkers etc
- **Knowledge base applications** : semantic web, expert systems, reasoning, medical diagnosis
- **Machine learning**: face recognition , autonomous cars, social network analysis, RS, detection systems

★ Hard or Strong AI

- AI research aims to create AI that can **replicate human intelligence completely!**
- **Strong ai** → machine that **approaches/supersedes human intelligence**
- If it can do typical human tasks, it can apply a wide range of background knowledge and **has some degree of self-consciousness.**
- Strong AI aims to build machines whose overall **intellectual ability is indistinguishable from that of a human being.**

★ Soft or weak AI

- refers to the use of software to study or accomplish specific problem solving or reasoning tasks that **do not encompass (cover) the full range of human cognitive abilities.**
- does **not achieve self-awareness**
- it is merely an intelligent, **specific problem-solver.**

★ AI goals

- **Replicate human intelligence** : still a distant goal.
- **Solve knowledge intensive tasks.**
- Make an **intelligent connection** between **perception** and **action**.
- Enhance human-human, human-computer and computer to computer interaction / communication.

• Engineering based AI Goal

- Develop concepts, theory and practice of **building intelligent machines**
- **Emphasis is on system building.**

• Science based AI Goal

- Develop concepts, mechanisms and vocabulary to **understand biological intelligent behavior.**
- **Emphasis** is **on understanding** intelligent behavior.

★ Four schools of thoughts (Russel & Norvig)

	Human-like	Rational
Thinking	Thinking Humanly	Thinking Rationally
Acting	Acting Humanly	Acting Rationally

1 Thinking Humanly

- approach tries to make machines **think the same way humans think**
- focuses on:
 - **Cognitive science + Psychology + Neuroscience + Brain modeling**
- Focus is **not just on behavior and I/O**, but **looks at reasoning process**.
- Goal is not just to produce human-like behavior but to **produce a sequence of steps of the reasoning process, similar to the steps followed by a human** in **solving the same task**.

2 Thinking Rationally

- based on **formal logic**.
- What is the correct way to think logically?
- Uses: **Propositional logic + First-order logic + Mathematical reasoning**
- Focus is on **inference mechanisms** that are probably correct and guarantee an optimal solution.
- **Study of computations that make it possible to perceive, reason and act**

3 Acting Humanly

- approach **focuses on behavior, not internal thinking**.
- If it behaves like a human, it is intelligent.

📌 To pass the test, AI needs:

- **Natural language processing + Knowledge representation + Reasoning + Machine learning**

🔍 **Turing Test:**

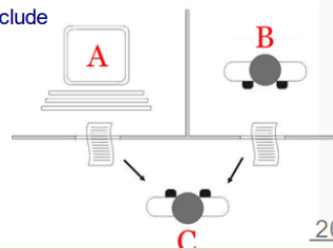
If a human talks to:

- A human
- A machine

And cannot tell which is which... 👉 The machine passes.

Example : Turing Test

- 3 rooms contain: a person, a computer, and an interrogator.
- The interrogator can communicate with the other 2 by teletype (to avoid the machine imitate the appearance or voice of the person).
- The interrogator tries to determine which is the person and which is the machine.
- The machine tries to fool the interrogator to believe that it is the human, and the person also tries to convince the interrogator that it is the human.
- If the machine succeeds in fooling the interrogator, then conclude that the machine is intelligent.
- Goal is to develop systems that are human-like.



e) * LIMITATIONS :-

• Technical:- Excludes physical interaction, only text-based conversations are evaluated, meaning it does not assess perception, physical reasoning. So, it measures conversational ability only rather than full artificial intelligence.

• Ethical:- the test emphasizes on deception rather than usefulness. system may pass by producing misleading responses / avoiding difficult questions. The focus on imitation may encourage development of systems designed to mislead rather than assist users transparently.

• Practical:- evaluation depends on human judges, which can be biased. different judges may reach different conclusions, ^{hence,} reducing ~~etc~~ reliability!

• Effect of on reliability & acceptance:-

↳ passing doesn't guarantee "true" intelligence.

↳ creates false perceptions of machine capability.

Therefore, turing test not considered a sufficient / reliable standalone measure of modern AI systems.

[limitations
of turing
test]

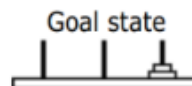
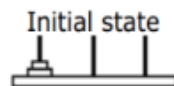
4 Acting Rationally

- Instead of copying humans or logic alone, it asks: **Does the agent act in a way that maximizes success?**
- Eg: A self-driving car
- Tries to **explain and emulate intelligent behavior in terms of computational processes**; that it is concerned with the **automation of intelligence**.
- Goal is to **develop systems** that are **rational and sufficient**.



Puzzle : Towers of Hanoi with only 2 disks

Solve the puzzle :

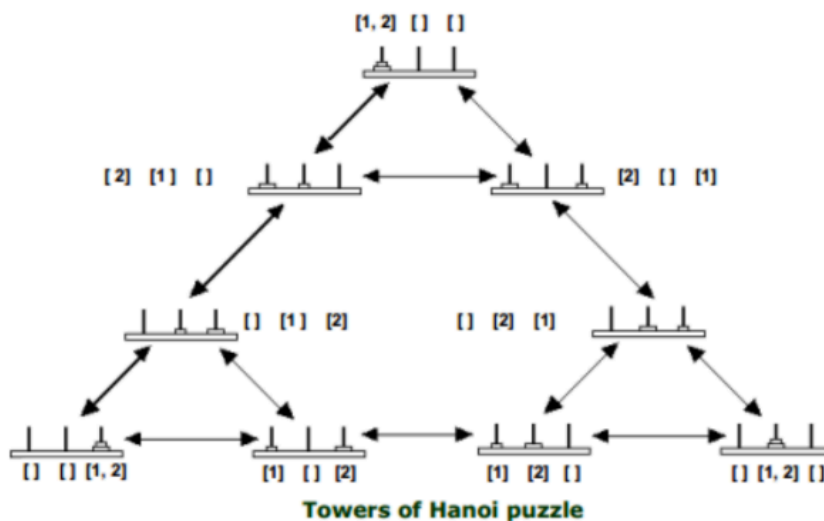


Move the disks from the leftmost post to the rightmost post while

- never putting a larger disk on top of a smaller one;
- move one disk at a time, from one peg to another;
- middle post can be used for intermediate storage.

Play the game in the smallest number of moves possible.

- Possible state transitions in the Towers of Hanoi puzzle with 2 disks.



- **Shortest solution** is the sequence of transitions from the top state downward to the lower left.

★ Important Concepts and Terms

♦ Agents

An **agent** is **anything that perceives** its **environment** **through sensors** and **acts upon it using actuators**.

♦ Automated reasoning

Automated reasoning is the **use of algorithms to derive logical conclusions automatically from given knowledge**.

♦ Cognitive science

Cognitive science is the interdisciplinary **study of how humans think, learn, perceive, and reason**.

♦ Computer science

Computer science is the **study of algorithms, computation**, and the **design** of computer systems.

♦ Intelligence

Intelligence is the **ability to learn, reason, adapt, and make decisions to achieve goals**.

♦ Intelligent agent

An **intelligent agent** is an agent that **acts rationally** to **achieve its goals based on its perceptions and knowledge**.

♦ Knowledge representation

Knowledge representation is the method of **encoding information about the world so a computer can reason with it**.

- ♦ **Linguistics**

Linguistics is the scientific **study of language structure, meaning, and communication.**

- ♦ **Lisp**

Lisp is one of the **earliest programming languages widely used in AI research** for symbolic processing.

- ♦ **Logic**

Logic is a **formal system of rules** used to **represent and reason about truth.**

- ♦ **Machine learning**

Machine learning is a branch of AI that enables systems to **learn patterns from data** and **improve performance without explicit programming.**

- ♦ **Natural language processing (NLP)**

Natural language processing is the field of AI that enables computers to **understand, interpret, and generate human language.**

- ♦ **Neural network**

A **neural network** is a **computational model inspired by the human brain**, used for **pattern recognition and learning.**

- ♦ **Predicate logic**

Predicate logic (first-order logic) is a formal system that expresses **statements about objects and their relationships.**

- ♦ **Propositional logic**

Propositional logic is a formal system where **statements are either true or false** and **combined using logical operators**.

- ♦ **Rational agent**

A **rational agent** is an agent that **selects actions that maximize expected performance** given its knowledge and percepts.

- ♦ **Rationality**

Rationality is the property of **choosing actions that maximize expected success** based on available information.

- ♦ **Turing test**

The **Turing test** is a test of machine intelligence where a machine is considered intelligent if a human cannot distinguish it from another human in conversation.

★ **Assignment 1**

What is the common definition of “AI”? Do you agree?

AI is the science of building machines that can perform tasks requiring human intelligence; yes, I agree because modern AI systems can learn, reason, and make decisions.

- **Do you know any AI application?**

Yes, examples include self-driving cars, virtual assistants (like Siri), recommendation systems (Netflix), and medical diagnosis systems.

- **Should artificial intelligence simulate natural intelligence?**

Not necessarily; AI should focus on solving problems effectively, even if it does not exactly mimic human thinking.

- **What are the criticisms of AI research? Do you agree?**

Criticisms include **high cost, ethical concerns, job displacement, and unrealistic expectations**; I agree that ethical regulation and responsible development are important.

- **What is the relation between AI and logic? AI and philosophy?**
Logic and philosophy?

AI uses **logic for reasoning and decision-making**; AI is connected to philosophy through **questions about mind, intelligence, and ethics**; logic is a branch of philosophy that **studies correct reasoning**.

- **Explain the meaning of logic? Reasoning? Ontology?**

Logic is a formal system of **rules for valid thinking**; **reasoning** is the process of **drawing conclusions** from information; **ontology** is the study or **representation of what exists and how entities are related**.

- **What is Natural Language Processing? And how is it related to AI?**

Natural Language Processing (NLP) is a field of AI that enables computers to understand and generate human language.

- **Why and how are probabilistic and statistical methods used in AI?**

They are used to handle uncertainty and incomplete information by modeling likelihoods and **making predictions based on data.**

- **What are the major research approaches/schools in AI? Which one do you think is more productive?**

The four major approaches are thinking humanly, thinking rationally, acting humanly, and acting rationally; the **rational agent approach is the most productive** because it focuses on **optimal decision-making.**